

SIH26033 - Direct Farmer-to-Consumer Digital Agri-Marketplace PRD

- [Product Requirements Document \(PRD\)](#)
 - [SIH26033 — Direct Farmer-to-Consumer Digital Agri-Marketplace](#)
 - [1. Executive Summary & Background](#)
 - [2. Problem Statement & Objective](#)
 - [3. Target Users & Pain Point Mapping](#)
 - [4. Proposed Solution & System Architecture Overview](#)
 - [5. Key Features & Functional Requirements](#)
 - [6. User Journey & Workflow](#)
 - [7. Tech Stack & System Design](#)
 - [8. Unique Value Proposition \(UVP\)](#)
 - [9. Success Metrics & KPIs](#)
 - [10. Scope & Roadmap](#)
 - [11. Hackathon Execution Timeline \(36 Hours\)](#)

Product Requirements Document (PRD)

SIH26033 — Direct Farmer-to-Consumer Digital Agri-Marketplace

Team Track: Smart India Hackathon 2026 | **Domain:** Agriculture, FoodTech & Rural Development **Prototype**
Duration: 36 Hours

1. Executive Summary & Background

India's agricultural supply chain is one of the most fragmented in the world. Between the farmer's field and the consumer's plate, produce typically passes through **4-6 intermediaries** — village aggregators, commission agents (arhatiyas), wholesalers, mandi traders, and retailers. Each layer adds margin without adding proportional value, resulting in a structural imbalance where:

- Farmers realize only **30-35%** of the final consumer price for perishables (as per multiple NABARD/ICAR studies).
- Consumers pay **2-3x** the farm-gate price due to cumulative margins and logistics inefficiency.
- Price discovery remains opaque, leaving farmers vulnerable to distress selling, especially for perishable crops with short shelf lives.

Why this matters now: With smartphone penetration crossing 750M+ in India and UPI becoming the default payment rail even in rural markets, the infrastructure for disintermediation already exists — what's missing is a **trusted, AI-assisted marketplace layer** that connects supply directly to demand while solving the two hardest problems in agri-tech: **logistics-last-mile** and **trust/verification** between strangers transacting high-value perishable goods.

Our solution — a **Direct Farmer-Buyer Digital Marketplace** — addresses this by combining e-commerce style listing/discovery with AI-driven fair pricing, verified buyer networks, and integrated logistics tracking, creating a transparent, efficient alternative to the traditional mandi system without replacing farmers' ability to still access mandis if they choose.

2. Problem Statement & Objective

2.1 Formal Problem Breakdown

Root Cause	Resulting Effect
Multiple non-value-adding intermediaries in the supply chain	Reduced farmer margins, inflated consumer prices
Lack of real-time price transparency	Farmers unable to negotiate fairly; distress selling
No verified buyer network for direct trade	Trust deficit prevents farmers from bypassing known local traders
Fragmented/absent logistics for small-lot produce	High spoilage, farmers forced to sell locally at low prices
No demand forecasting at farm level	Overproduction/underproduction cycles, price crashes

2.2 Primary Goals

1. **Eliminate unnecessary intermediary layers** by enabling direct digital transactions between farmers/FPOs and consumers/bulk buyers.
2. **Increase farmer realization price** by providing AI-driven fair price recommendations benchmarked against real mandi and market data.
3. **Reduce consumer purchase price** by cutting out margin-stacking middlemen while preserving quality assurance.
4. **Build a trust layer** via buyer verification and rating systems to de-risk direct transactions for farmers unfamiliar with digital trade.
5. **Optimize logistics** using route optimization to reduce spoilage and delivery cost for low-volume, distributed farm produce.

2.3 Quantitative Target Outcomes (Prototype-Stage Simulation Targets)

Metric	Baseline (Current System)	Target (Our Platform)
Farmer share of consumer price	~30-35%	55-65%
Consumer price reduction	—	15-25% lower than retail market rate
Price discovery time	Manual, mandi-visit dependent	< 30 seconds (in-app)
Intermediary layers	4-6	1-2 (platform + logistics partner)
Order-to-delivery logistics efficiency (route optimization)	Ad-hoc	20-30% reduction in transit distance/time

3. Target Users & Pain Point Mapping

User Persona	Core Pain Point	How Our Solution Solves It
Small & Marginal Farmer (Primary)	Low bargaining power, forced distress selling, no price visibility, exploitation by middlemen	Direct listing to verified buyers + AI fair-price recommendation + real-time mandi price comparison dashboard
FPO (Farmer Producer Organization)	Aggregation and bulk-selling inefficiency, lack of digital sales channel	Bulk listing tools, aggregated order management, institutional buyer access
Consumer / Retail Buyer (Secondary)	High retail prices, uncertainty on freshness/source	Direct-from-farm pricing, transparency on produce origin, lower cost via disintermediation
Wholesalers / Institutional Buyers	Inconsistent supply, lack of verified sourcing network	Verified farmer/FPO listings, demand forecasting to plan procurement, trust-scored suppliers
Restaurants / Processors	Need reliable, quality-consistent bulk sourcing	Direct bulk ordering with quality/quantity filters and trust-score visibility

Logistics Providers	Fragmented, low-volume pickup routes are unprofitable	AI route optimization batches nearby orders into efficient multi-pickup routes
Government / Agri Officials	No real-time visibility into market linkage, price trends, farmer distress signals	Aggregated dashboard for market-linkage insights, price trend monitoring, and intervention triggers

4. Proposed Solution & System Architecture Overview

4.1 High-Level Conceptual Flow

The platform operates as a **three-sided marketplace** (Farmers/FPOs ↔ Platform Intelligence Layer ↔ Buyers) underpinned by a logistics coordination layer and an AI recommendation engine that touches both pricing and delivery routing.

Functional Pipeline: 1. Farmer/FPO onboards → lists crop (type, quantity, quality grade, location, expected availability date). 2. AI Pricing Engine ingests listing + historical mandi data + nearby demand signals → suggests fair price range. 3. Listing goes live on marketplace → visible to consumers/bulk buyers within relevant geographic/demand radius. 4. Buyer discovers listing → views trust score, price comparison, quality info → places order. 5. Order Management module triggers logistics workflow → route optimization engine batches pickup/delivery. 6. Payment processed (UPI/Razorpay) → held in escrow-like status until delivery confirmation. 7. Transaction completes → both parties rate each other → trust scores updated → data feeds back into AI demand forecasting model.

4.2 System Architecture (Mermaid Flowchart)

```
flowchart TD
    A[Farmer / FPO App] -->|List Crop: Type, Qty, Quality, Location| B[Backend API - FastAPI]
    C[Consumer / Bulk Buyer App] -->|Browse & Search| B
    B --> D[(PostgreSQL Database)]
    B --> E[AI Pricing Engine - Scikit-learn]
    E -->|Fair Price Suggestion| B
    B --> F[Demand Forecasting Module]
    F -->|Historical + Real-time Demand Signals| E
    B --> G[Buyer Verification & Trust Score Service]
    C -->|Place Order| B
    B --> H[Order & Logistics Management Service]
    H --> I[Route Optimization Engine - Google Maps/OSM API]
    I -->|Optimized Pickup/Delivery Route| J[Logistics Partner App]
    H --> K[Payment Gateway - UPI/Razorpay]
    B --> L[Government/Agri Official Dashboard]
    D --> E
    D --> F
    D --> L
    B -.JWT/Firebase Auth. .-> M[Auth Service]
```

4.3 Core System Layers

- **Presentation Layer:** Role-based dashboards (Farmer, Buyer, Logistics Partner, Govt Official) built as responsive web app.
- **Application/API Layer:** FastAPI-driven REST services handling listings, orders, verification, and transactions.
- **Intelligence Layer:** ML models for fair-price recommendation and demand forecasting, plus route optimization logic.
- **Data Layer:** PostgreSQL for transactional and relational data (users, listings, orders, price history).
- **Integration Layer:** Maps API for geolocation/routing, Payment Gateway for transactions, Firebase/JWT for auth.

5. Key Features & Functional Requirements

5.1 Feature Breakdown with Sub-Components

- Feature 1: Direct Farmer-Buyer Marketplace** - Crop listing creation (type, variety, quantity, quality grade, harvest/availability date, location) - Search & filter for buyers (crop type, location radius, price range, quantity) - Bulk listing support for FPOs - Listing status tracking (active, sold, expired)
- Feature 2: Price Discovery & Comparison Dashboard** - Real-time/nearby mandi price feed integration (scraped or API-based) - Side-by-side comparison: mandi price vs. buyer offers vs. platform-recommended price - Historical price trend graphs per crop/region
- Feature 3: AI-Based Fair Price Recommendation** - Regression model trained on crop type, quantity, quality, location, seasonality, historical mandi prices - Confidence range output (min-max fair price) rather than single point estimate - Continuous learning loop from completed transaction data
- Feature 4: Buyer Verification & Trust Score** - KYC-lite onboarding (phone/Aadhaar-linked verification, business registration for bulk buyers) - Trust score computed from: transaction completion rate, payment punctuality, farmer ratings - Visible badge system (e.g., Verified Buyer, Top Rated) on buyer profiles
- Feature 5: Logistics & Order Management** - Order lifecycle tracking: placed → confirmed → picked up → in-transit → delivered → paid - Route optimization for logistics partners batching multiple nearby pickups - Payment status tracking with UPI/Razorpay integration - Delivery confirmation via OTP/photo proof

5.2 Must-Have (MVP) vs. Nice-to-Have (Phase 2)

Feature Component	Priority
Farmer/Buyer registration & role-based login (JWT/Firebase)	Must-Have (MVP)
Crop listing creation & browsing	Must-Have (MVP)
Basic price comparison (mandi price vs. listed price)	Must-Have (MVP)
AI fair-price recommendation (baseline regression model)	Must-Have (MVP)
Order placement & status tracking	Must-Have (MVP)
Basic trust score (rating-based only)	Must-Have (MVP)
UPI/Razorpay payment integration	Must-Have (MVP) <i>(can be mocked if time-constrained)</i>
Route optimization (basic distance-based)	Must-Have (MVP) <i>(simplified version)</i>
Government/Agri official monitoring dashboard	Nice-to-Have (Phase 2)
Advanced demand forecasting (time-series model)	Nice-to-Have (Phase 2)
Full KYC/Aadhaar-based buyer verification	Nice-to-Have (Phase 2)
Multi-language/voice interface for farmers	Nice-to-Have (Phase 2)
Escrow-based payment holding	Nice-to-Have (Phase 2)
SMS/WhatsApp-based listing for low-smartphone-literacy farmers	Nice-to-Have (Phase 2)
Insurance/quality dispute resolution module	Nice-to-Have (Phase 2)

6. User Journey & Workflow

6.1 Farmer Journey

- Onboarding:** Farmer registers via phone number (OTP-based auth) → selects language → completes basic

profile (location, crop types grown).

2. **Listing Creation:** Farmer/FPO lists crop with quantity, quality grade, expected harvest date, and location (auto-geotagged).
3. **Price Guidance:** AI engine instantly displays a recommended fair price range alongside nearby mandi rates.
4. **Publish:** Listing goes live, visible to buyers within relevant radius/demand zones.
5. **Order Receipt:** Farmer receives order notification with buyer trust score visible; can accept/negotiate.
6. **Logistics Handoff:** Pickup scheduled via optimized route; farmer gets pickup time window.
7. **Payment & Closure:** Payment confirmation received post-delivery; farmer rates buyer.

6.2 Buyer Journey

1. **Onboarding:** Buyer registers, completes basic verification (business type for bulk buyers).
2. **Discovery:** Browses/searches listings by crop, location, price, quantity.
3. **Comparison:** Views price comparison dashboard and farmer/FPO trust score.
4. **Order Placement:** Places order, selects delivery preference, makes payment.
5. **Tracking:** Tracks order status through logistics pipeline in real time.
6. **Delivery & Rating:** Confirms receipt, rates farmer/FPO.

6.3 Government/Agri Official Journey (Phase 2)

1. Logs into monitoring dashboard.
2. Views aggregated price trends, transaction volumes, and regional demand-supply gaps.
3. Flags anomalies (e.g., price crashes) for potential intervention.

7. Tech Stack & System Design

Layer	Technology	Selection Justification
Frontend	React.js + Tailwind CSS	Fast to prototype, component-reusability across farmer/buyer/logistics dashboards, responsive out-of-the-box for mobile-first rural users
Backend	Python + FastAPI	High-performance async API framework, native compatibility with ML model serving (same language as AI/ML stack), auto-generated OpenAPI docs speed up hackathon integration
Database	PostgreSQL	Relational integrity needed for orders/transactions/users; strong geo-query support (PostGIS-compatible) for location-based listing search
Authentication	JWT / Firebase Authentication	Firebase enables quick phone-OTP auth critical for rural farmer onboarding; JWT secures backend API sessions
AI/ML	Python, Scikit-learn	Rapid model prototyping for regression-based fair-price prediction within hackathon timeframe; lightweight enough for real-time inference
Maps & Location	Google Maps API / OpenStreetMap	Enables geolocation-based listing discovery and route optimization for logistics batching; OSM as free fallback
Cloud	AWS / Firebase	Scalable hosting, Firebase simplifies real-time sync for notifications and auth; AWS for backend/DB hosting flexibility

Payment	UPI / Razorpay API	UPI is the dominant rural-India payment rail; Razorpay provides quick sandbox integration for prototype demo
---------	--------------------	--

8. Unique Value Proposition (UVP)

Dimension	Existing Alternatives (eNAM, local mandi apps, generic agri e-commerce)	Our Solution
Intermediary Removal	eNAM digitizes mandi trading but still routes through licensed mandi traders	True peer-to-peer listing directly to consumers/bulk buyers, bypassing mandi trader layer entirely
Price Intelligence	Static price boards, no personalized recommendation	AI-driven fair price recommendation personalized to crop quality, quantity, and location — not just a generic mandi rate
Trust Mechanism	Little to no buyer-side accountability for direct trade	Built-in trust score + verification system designed specifically to de-risk stranger-to-stranger agri transactions
Logistics Integration	Most platforms leave logistics to the farmer/buyer to arrange independently	Integrated route optimization batching nearby orders for cost-effective last-mile logistics
User Accessibility	Often complex, trader-oriented UX	Mobile-first, simple listing flow designed for low digital literacy, with regional language support roadmap
Feedback Loop	No continuous learning from transaction outcomes	AI models improve over time using completed transaction data for better forecasting and pricing accuracy

9. Success Metrics & KPIs

9.1 Technical KPIs

- **API response latency:** < 300ms for listing/search queries
- **AI price recommendation inference time:** < 1 second
- **Model accuracy (fair price prediction):** ≥ 80% within ±10% of eventual transaction price (validated against historical mandi data)
- **System uptime (demo/prototype):** 99%+ during evaluation window
- **Route optimization improvement:** 20–30% reduction in simulated transit distance vs. naive routing

9.2 Business / Social Impact KPIs

- **Farmer price realization uplift:** Target 55–65% of consumer price (vs. ~30–35% baseline)
- **Consumer price reduction:** 15–25% lower than prevailing retail rate for equivalent produce
- **Number of active farmer/FPO listings** (demo-stage: simulated dataset of 50–100 listings)
- **Buyer trust score adoption:** % of transactions completed with verified buyers
- **Order completion rate:** % of placed orders successfully delivered and paid

10. Scope & Roadmap

10.1 In-Scope (Hackathon MVP — 36 Hours)

- Farmer & Buyer registration/login (JWT/Firebase OTP-based)
- Crop listing creation with quantity, quality, location, price
- AI-based fair price recommendation (baseline regression model on sample/synthetic mandi dataset)
- Price comparison dashboard (mandi price vs. platform price)
- Order placement and basic status tracking (placed → confirmed → delivered)
- Basic trust score (rating-based, no full KYC)
- Simplified route optimization (distance-based batching demo using Maps API)
- Mocked/sandbox UPI-Razorpay payment flow
- Functional demo across Farmer, Buyer, and (if time permits) Logistics dashboards

10.2 Out-of-Scope (Post-Hackathon / Future Production Scope)

- Full Aadhaar-based KYC verification pipeline
- Advanced time-series demand forecasting with real-time market data ingestion
- Government/Agri official analytics dashboard with intervention alerts
- Multi-language voice/IVR interface for low-literacy farmers
- Escrow-based payment holding and dispute resolution
- SMS/WhatsApp-based listing fallback for non-smartphone users
- Insurance and produce-quality dispute mediation module
- Integration with real eNAM/state mandi APIs for live price feeds
- Scalable multi-region deployment with load balancing and caching layers
- Cold-chain/refrigerated logistics partner integration

11. Hackathon Execution Timeline (36 Hours)

Time Block	Milestone	Key Activities
Hour 0-4	Kickoff & Planning	Finalize architecture, split tasks by role, set up repos/environments, design DB schema
Hour 4-10	Core Backend Setup	FastAPI project scaffolding, PostgreSQL schema implementation, auth (JWT/Firebase) setup
Hour 10-16	Core Frontend + Listing Flow	React UI for farmer listing creation, buyer browsing/search, basic dashboards
Hour 16-22	AI Module Integration	Train baseline Scikit-learn price recommendation model on synthetic/historical mandi dataset, integrate inference API
Hour 22-26	Order & Logistics Flow	Order management APIs, basic route optimization using Maps API, order status UI
Hour 26-30	Trust Score + Payment Integration	Implement rating-based trust score, mock/sandbox UPI-Razorpay flow
Hour 30-33	Integration & Testing	End-to-end flow testing (farmer list → buyer order → payment → delivery), bug fixes
Hour 33-35	UI Polish & Deployment	Final UI/UX polish, deploy to AWS/Firebase hosting, prepare demo dataset
Hour 35-36	Pitch Prep	Finalize slide deck, rehearse demo flow, prepare Q&A responses for judges

This PRD is structured for modular reuse — each section maps directly to a corresponding slide in the SIH pitch deck (Problem → Solution → Architecture → Features → Tech Stack → Impact → Roadmap).